Def. Let X be a Metric space. Denote $B(X) \overset{\text{def}}{=} \{f : X \to \mathbb{R} \mid f \text{ is bounded}\}$.

In this page, unless otherwise stated, we assume that $\sup = \sup\limits_{x \in X}$.

Prop. Given $f, g \in B(X)$,

$$|\sup f - \sup g| \leq \sup|f - g| \leq \sup|f| + \sup|g|.$$

and

$$\sup(f+g) \leq \sup f + \sup g.$$